Sentence Level Correction of Common Spoken Grammatical Errors with Deep Neural Networks

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***Abstract*— Grammatical errors disrupt conversation clarity and effectiveness, even for fluent speakers. Such errors can make even those fluent in a language oblivious to what is being conveyed. This study presents a novel approach, using natural language processing and transformer models to both identify and provide feedback on common grammatical errors spanning 10 types. These include adjective and adverb misuse, determiner selection, preposition selection, conjunction selection, verb form and tense, subject-verb agreement, noun and pronoun errors.**

The system operates as a cascaded pipeline of a binary sentence classifier and a grammar corrector. Sentences are classified as "correct" or "incorrect" by the classifier while Google's T5 model, fine-tuned on C4-200M for grammatical error correction (GEC), corrects errors in ungrammatical sentences. ERRANT annotates the errors present in the input sentence, concluding the evaluation. Model validation performed on various datasets shows the grammar corrector performs best on error-filtered JFLEG with BLEU 0.89 and ROUGE 0.95. The binary sentence classifier also achieved an accuracy score of 0.78 on a custom dataset. While it is effective, its applicability is currently limited to English, with plans for multilingual support and real-time applications exploring the expansion of the model's capabilities.

Keywords*— *Grammar error correction, deep learning, spoken errors, sentence classification, grammar error detection

I. INTRODUCTION

Grammatical inaccuracies in spontaneous speech can disrupt communication, making it difficult for listeners to comprehend the intended message. Mistakes in grammar can significantly reduce clarity and disrupt the flow of conversation. The more language technology such as virtual assistants, automated transcription services, and interactive voice systems become part of everyday life, the greater the need to automatically correct spoken grammatical errors. Speech-to-text systems will become more accurate if the technologies that can recognize and fix such problems automatically are developed. Real-time detection and correction of the typical grammar mistakes can be a means of clear communication, especially in circumstances when spoken instructions or replies are crucial. Furthermore, manual grammar error detection (GED) and grammar error correction (GEC) are both time-consuming and impracticable. For this reason, advanced automated systems that can perform this work effectively are required.

This study aims to develop a system that is automated to correct common grammatical mistakes in spoken English, thereby enhancing communication clarity and efficacy. Ten particular grammatical faults that are commonly seen in speech are the focus of this investigation. These errors are adjective and adverb misuse, determiner selection, preposition selection, conjunction selection, verb form and tense, subject-verb agreement, noun and pronoun errors. As they occur frequently, they serve as an impactful target for correction. Concentrating on frequent mistakes helps the system to streamline error correction and offer valuable feedback to users, thus aiding in their language skill development.

The structure of the remainder of this paper is as follows: Section 2 accommodates literature survey covering existing approaches to grammatical error detection and correction, focusing on recent advances in GED and GEC. Section 3 outlines methodology, including the data preparation process and the architecture of the proposed system. The implementation details, covering model training and evaluation, are discussed in Section 4. Section 5 shows the results of the experiments, followed by an analysis of the system's performance. Finally, Section 6 talks about the conclusion of the paper and potential discussions in directions for future work.

II. PREVIOUS WORK

The literature on GED and GEC for English has evolved significantly, highlighting the importance of automated approaches in enhancing communication clarity. Early research emphasised the role of error-correction detection and response generation in spoken dialogue systems, showcasing foundational techniques for identifying and correcting speech errors in real time as mentioned by Bulyko et al [1]. Some of the other earlier approaches include an n-gram language model [2] for grammatical feedback in spoken English, and a

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rule-based model for correcting grammar in academic writing settings [3]

Recent studies have adopted more advanced models, such as LSTMs, demonstrating improved accuracy in identifying grammatical issues within conversational contexts as shown by Huang, 2023 [4]. A study performed by Chen at al [5] demonstrates that an English GEC model that combines the attention mechanism with recurrent neural networks (RNN) enhances accuracy by using a bidirectional gated recurrent network in a dual encoder structure. Zhu et al [6] performed a comparative study of three different architectures for the purpose of seq2seq error detection and correction, namely - LSTM, transformer and BERT. The experimental results include F-scores of 83%, 84% and 88%, respectively. In a similar vein, He [7] employed LSTMs and transformers to perform error detection in English composition, resulting in F-scores of 0.3599%, 0.3358% and 0.7725% for verb errors, subject verb consistency errors and overall detection, respectively. Among the transformer-based approaches, T5 models were fine-tuned for real-time grammatical error correction in spoken language [8], with T5-Base and T5-Small achieving accuracies of 0.60 and 0.56 respectively . Another study [9] explored BART for the GEC task, showing competitive results. These findings highlight the effectiveness of transformer models, especially T5, in addressing grammatical error correction, with T5 proving particularly well-suited for both written and spoken language correction tasks.

In order to sharpen focus and enhance feasibility, a careful analysis of commonly occurring errors is thus of paramount importance. According to Ting et al [10], prepositions are the most difficult for non-native speakers/students, constituting about 35% of total errors, followed by word form and articles. It has also been found that a mismatch between pronoun and noun usage occurs at a frequency of 43.1% [11], while Tanashur et al [12] mentioned that omission of subject verb agreement occurred 16.29% of the time. Some of the most commonly occurring errors as mentioned by Nangle et al [13] are article related errors, word choice errors, pronoun errors, preposition errors, grammar tense errors and noun/pluralization errors. Damaiyanti [14] mentioned that over generalisation is a common form of error and occurred in 14% of the results. As for tenses, the most relevant errors in verb tenses include mixed formation error at 62.50%, omission of word error at 25%, missed ordering error at 7.14% and addition error at 5.36% [https://doi.org/10.21107/prosodi.v13i2.6203]

GEC can be modelled as a neural machine translation task [16] [17], where ungrammatical English is considered the source and correct, grammatical English is the target. Thus, as part of the evaluation metrics, this study uses the BLEU (Bilingual Evaluation Understudy) method [18] and ROUGE (Recall-Oriented Understudy for Gisting Evaluation) method [19] for assessing the quality of corrections generated. BLEU evaluates translation accuracy by systematically comparing machine-generated translations with human reference translations, calculating the score based on this comparison. On the other hand, ROUGE compares the overlap of n-grams, word sequences and word pairs between the generated output and reference summaries focusing more on recall. Together, BLEU and ROUGE provide a comprehensive evaluation framework, addressing both precision and recall in language generation tasks. These approaches marked a major breakthrough in establishing standardised techniques for evaluating machine translation quality and accuracy.

III. METHODOLOGY

The proposed system consists of three main phases: GED, GEC and error annotation. They work in tandem, with information cascading from the detection model to the correction model to finally, the annotator. The expected input is a speech transcript, tokenized into sentences, since all models work at the sentence level.

In phase I, the GED model, or the RoBERTa-based sentence classifier, works by analysing each sentence, detecting grammatical errors (if any) and consequently flagging sentences that require correction. The flagged sentences move on to phase II, the t5-based grammar corrector. The model generates the grammatically correct version of each sentence that is given as input.

Finally, the nature of errors corrected is obtained by employing ERRANT (ERRor ANnotation Toolkit) [20] in phase III. ERRANT is an annotation tool which utilises linguistic rules and parsing techniques to analyse and classify edits between two parallel sentences. It uses the M2 format to align original and corrected sentences, part-of-speech (POS) tagger and dependency parser to understand the syntactic structure of sentences. Thus, it accepts two sentences as input - the original, erroneous sentence and the corrected version of the same sentence as produced by the grammar corrector. Upon analysing both, ERRANT labels each error occurrence with POS type and one the following prefixes: “M” for missing, “R” for replacement or “U” for unnecessary.

```
INPUT: PARALLEL SENTENCES
Original sentence:  The team been working hard at the project
Corrected sentence:  The team has been working hard on the project

OUTPUT: ERROR ANNOTATIONS
1. [NULL]  --->   has  | Error type:  M:VERB:TENSE
2. at      --->   on   | Error type:  R:PREP
```

Fig. 1. Example of the input given to ERRANT, followed by output produced. The first error is a verb tense error, where the word “has” is missing (M). The second error is the wrong usage of “at”, which replaced the correct preposition “or” in the original sentence.

To streamline the model’s focus, the ten most frequent errors in spoken English have been taken into consideration while training the sentence classifier and the grammar corrector. These include the incorrect usage of adjective forms, adverbs, determiners, conjunction, preposition, verb forms, subject verb agreement, verb tense, noun and pronouns. This is reflected in the training dataset by filtering it to retain only these particular instances of errors. A detailed overview of the filtering process can be found in section IV.

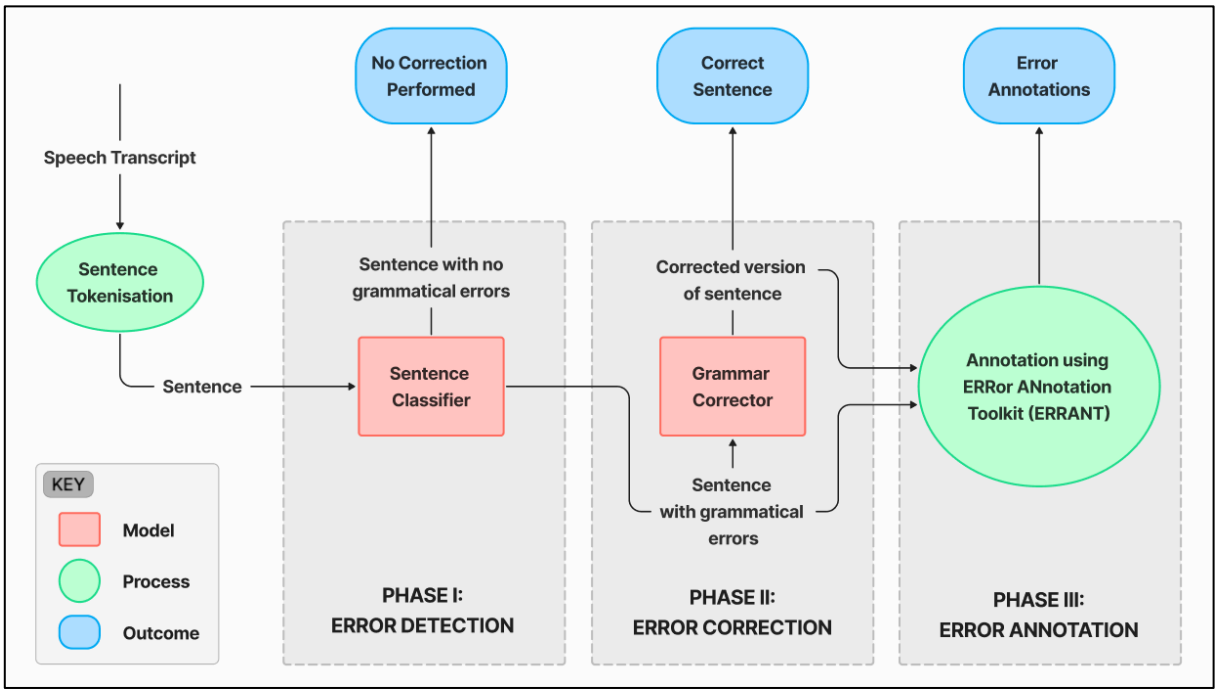


Fig. 2. High level view of the working of the system, beginning from speech transcript to final result containing error annotations

IV. IMPLEMENTATION

A. Dataset

The C4-200M[1] grammar error correction synthetic dataset is a large scale corpus consisting of 185 million English sentence pairs, created using a Tagged Corruption model on Google's C4 dataset. The dataset comprises a pair of grammatically erroneous sentences and its grammatically correct equivalent. The dataset has an extensive coverage making it highly suitable for grammar error correction.

B. Preprocessing

For this study, a sample of 90 million sentence pairs of the C4-200M dataset was used. The sample dataset was processed using the ERRANT tool to automatically annotate parallel English sentences with error type information. The data samples from C4-200M which exhibited the error of interest, as listed in Fig. 3, were retained. The filtering process reduced the dataset to 12 million entries ensuring that only the relevant error labelled instances were used for model training and evaluation.

The sentence classifier required binary labelled data, for which the structuring of C4-200M was altered slightly. To prepare the data for classification, a binary label was assigned to every sentence in each pair belonging to the original dataset. Sentences that are grammatically incorrect were labelled as 0, while the grammatically correct sentences were labelled as 1. The modified labelled dataset was then used for the GED task in phase 1, which aims to determine if the input sentence necessitates grammatical correction.

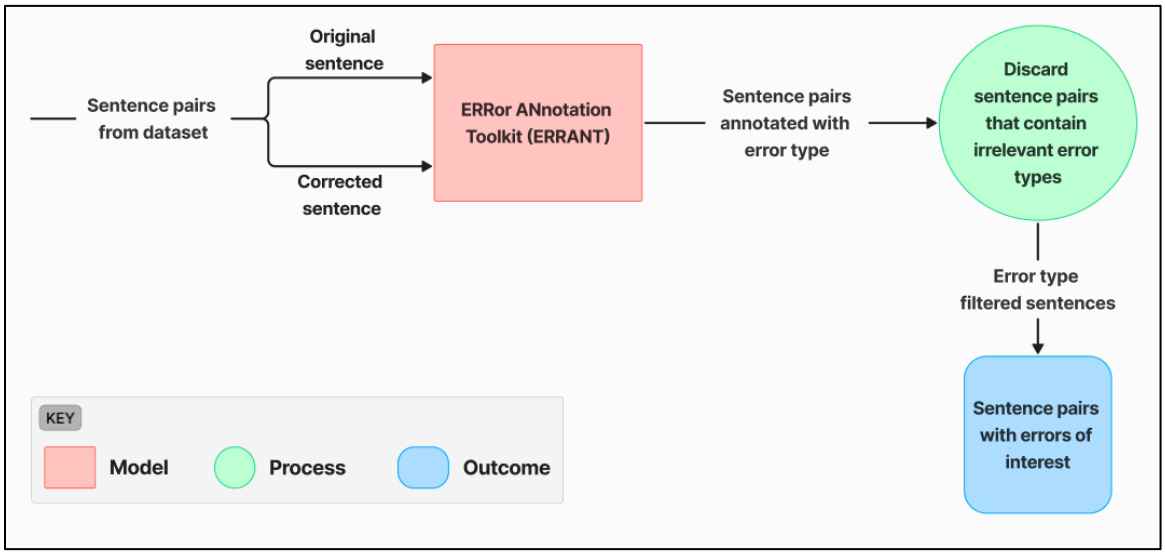


Fig. 3. Preprocessing method used to discard unnecessary errors

TABLE I. ERRRANT ANNOTATION CODES

ANNOTATION CODE	ERROR TYPE
ADJ:FORM	Adjective Form
ADV	Adverb
DET	Determiner
CONJ	Conjunction
PREP	Preposition
VERB:FORM	Verb Form
VERB:SVA	Subject Verb Agreement
VERB:TENSE	Verb Tense
NOUN	Noun
PRON	Pronoun

C. *Sentence Classifier*

The sentence classifier is based on the RoBERTa base model, a transformer-based architecture designed for natural language processing tasks. It was developed by researchers at Facebook AI in 2019 is an extension of BERT[] that optimises its training methodology. The model is initialised from pre-trained weights (roberta-base) which have been trained on a large corpus of text (including web pages and books) using masked language modelling to understand contextual language representations.

For this specific task, the RoBERTa base model was fine-tuned for binary classification designed to check whether an input sequence matches the correct grammatical label. The fine-tuned roberta-base contains 12 transformer layers each of which uses self-attention mechanisms to capture contextual relationships between tokens followed by classification heads. The adjusted hyperparameters of the model includes a learning rate of 5e-5, batch size of 64 and 3 epochs of training. The training process is optimised using AdamW to adjust the learning rate, ensuring rapid convergence and robustness.

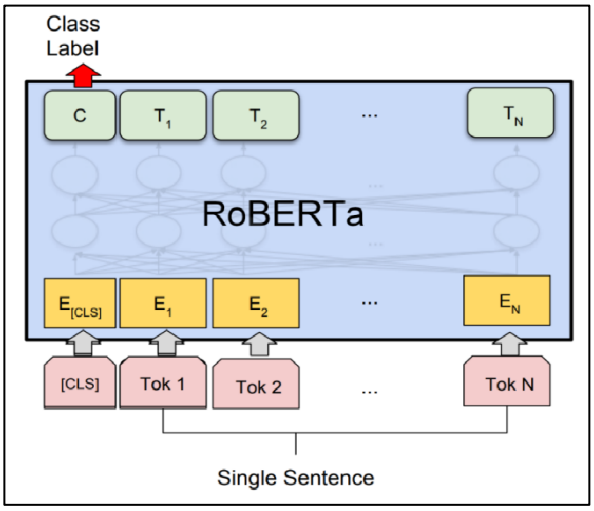


Fig 4. High level architecture of RoBERTa[2]

D. *Grammar Corrector*

The grammar corrector, based on Google's T5 (Text-to-Text Transfer Transformer), was fine-tuned to perform seq2seq correction by generating the correct version of the erroneous sentence. T5 is generally used for text transformation tasks, taking input as text and giving output in text form. It utilises an encoder-decoder architecture which has a series of transformer blocks where each block has a multi-head self-attention mechanism, a feed-forward network and a layer normalisation step. T5-base, which has 220M parameters, was used for grammar correction tasks. It consists of 12 encoder layers, 12 decoder layers, with a hidden representation dimensionality of 768, 12 attention heads per layer, and a feed-forward layer size of 3072.

Owing to the significant size of the dataset, it was not possible to load its entirety to main memory in a single shot. Thus, the fine tuning process involved splitting the data into chunks, where each chunk contained 100000 sentence pairs. The dataset was loaded from secondary memory chunk by chunk, resulting in training that took place in batches. During training, each batch of input was tokenized, padded and passed to the grammar corrector, which learnt the semantic difference between incorrect and corrected text.

The model was optimised using AdamW optimizer with a learning rate of 5e-5 during gradient backpropagation. The training process iterates for 1 epoch, where the model processes each batch of data, computes the loss for the same and adjusts its parameters to minimise this loss.

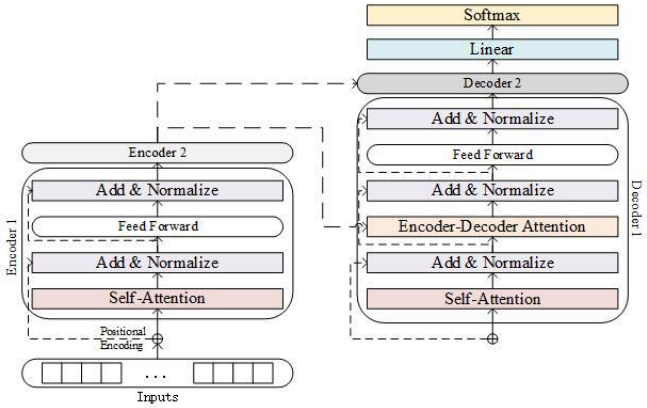


Fig 4. High level architecture of T5[3]

E. *Experimental Setup*

The model training setup utilised an Nvidia RTX4090 GPU with 24GB VRAM and a system with 128GB RAM, running on Windows 10. The model was developed and trained using VS Code as the IDE.

V. RESULTS

In this section, the performance results of both the models on previously unseen data are presented. The evaluation aims to demonstrate the effectiveness of these models across multiple datasets and validation techniques.

A. *Sentence Classifier - Validation*

The evaluation of the sentence classifier was performed by dividing the data into training and test sets using an 80-20 ratio. The model was assessed on multiple metrics including accuracy, precision, recall and F1-score. It achieved an

accuracy of 0.78, demonstrating the model's ability to distinguish between correct and incorrect sentences with minimal false positives and false negatives.

B. Grammar Corrector - Validation

The grammar corrector was evaluated on several datasets that were split into two categories: filtered and unfiltered. The filtered datasets contains sentences with only targeted grammatical errors, while the unfiltered datasets represented real world data that included other errors too. As expected, the results illustrated that the model performed better on filtered datasets, where error types were more controlled, thus achieving higher BLEU and ROUGE scores of 0.89 and 0.95. In contrast, the performance on unfiltered datasets was slightly lower, BLEU score 0.75 and ROUGE score of 0.89, but still competitive, reflecting the model's capacity to handle diverse errors.

TABLE II. PERFORMANCE METRICS OF THE SENTENCE CLASSIFIER ON THE CUSTOM C4-200M BASED DATASET

Class	Precision	Recall	F-Score
Incorrect	0.84	0.69	0.76
Correct	0.84	0.84	0.84
Accuracy			0.78
Macro avg.	0.79	0.78	0.78
Weighted wvg.	0.79	0.78	0.78

TABLE III. PERFORMANCE METRICS OF THE GRAMMAR CORRECTOR ON UNFILTERED DATASETS

Dataset	BLEU	Rouge1	Rouge2	RougeL	RougeLSum
BEA_2019	0.75	0.90	0.81	0.89	0.89
JFLEG	0.64	0.85	0.71	0.84	0.84
PIE_Synthetic	0.75	0.90	0.88	0.90	0.90

TABLE IV. PERFORMANCE METRICS OF THE GRAMMAR CORRECTOR ON FILTERED DATASETS

Dataset	BLEU	Rouge1	Rouge2	RougeL	RougeLSum
C4-200M	0.85	0.93	0.87	0.93	0.93
BEA_2019	0.84	0.92	0.84	0.92	0.92
JFLEG	0.89	0.95	0.89	0.95	0.95

VI. CONCLUSIONS AND FUTURE SCOPE

In an attempt to simplify and automate grammatical error correction in the context of spoken English, the proposed system performs error detection followed by correction, finally concluding with annotation with the help of ERRANT. Focussing on ten most relevant errors - adjective forms, adverbs, determiners, conjunction, preposition, verb forms, subject verb agreement, verb tense, noun and pronouns - the proposed approach provides the correct sentence, along with the type of error, which can be either missing, replacement or unnecessary. We have achieved significant results on held out data from C4-200M as well other benchmark datasets such as BEA2019 and JFLEG.

While the system performs well in grammar correction, its functionality is currently limited to English, restricting its global usability. Future research may enhance the model's ability to work with more languages and integrate it with real-time applications. Additionally, hybrid model integration with complementary approaches can be explored for more nuanced grammar correction. Finally, Exploring hybrid systems that integrate neural networks with rule-based approaches could further enhance the accuracy and adaptability of real-time grammar correction in spoken languages.

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